

Outline:

- Introduction to Kubernetes
- Exercises on the different resources: probes, HPA and Jobs

Study environment:

- Lab materials are present on Moodle
- We will use AWS Academy resources:
 - https://www.awsacademy.com/vforcesite/LMS_Login
- In our laboratory, we will use :
 - aws-cli - <https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2-linux.html>
 - kubectl - <https://kubernetes.io/docs/tasks/tools/>

When you execute all commands from command line with version argument they should not exit with error

Note: If you are using desktops in Multimedia Laboratory, please select Ubuntu image (credentials inf3/inf3). Due to issues with aws-cli version please execute following command:

In -s /snap/bin/aws .local/bin/aws

Restart your terminal, you should use aws-cli version 2+ (command: aws version)

- Please combine the results of all tasks in one single file and attach it in UPEL (at the end of class meeting).
- If you are using desktops, please make sure that the time is properly set
- At the end of lab:
 - Click the reset button on Vocareum labs

Kubernetes

Kubernetes allows the creation of different objects, including probes, jobs, and horizontal scaling policies.

Task -1 – Prepare and Verify your environment

First, using AWS Console create cluster in Elastic Kubernetes Service (in search dialog type EKS)

Step 1

Configure cluster

Step 2

Specify networking

Step 3

Configure observability

Step 4

Select add-ons

Step 5

Configure selected add-ons settings

Step 6

Review and create

Configure cluster

Configuration options - new Info

Choose how you would like to configure the cluster.

☐ Quick configuration (with EKS Auto Mode) - new
 Quickly create a cluster with production-grade default settings. The configuration uses EKS Auto Mode to automate infrastructure tasks like creating nodes and provisioning storage.

☒ Custom configuration
 To change default settings prior to creation, choose this option. This configuration gives the option to use EKS Auto Mode and customize the cluster's configuration.

EKS Auto Mode - new Info

Choose if you would like to use EKS's Auto Mode.

☒ Use EKS Auto Mode
 EKS automates routine cluster tasks for compute, storage, and networking. When a new pod can't fit onto existing nodes, EKS creates a new node. EKS combines cluster infrastructure managed by AWS with integrated Kubernetes capabilities to meet application compute needs. [View pricing](#)

► Included capabilities

Cluster configuration Info

Name
Enter a unique name for this cluster. This property cannot be changed after the cluster is created.

The cluster name should begin with letter or digit and can have any of the following characters: the set of Unicode letters, digits, hyphens and underscores. Maximum length of 100.

Cluster IAM role Info
Select the Cluster IAM role to allow the Kubernetes control plane to manage AWS resources on your behalf. This cannot be changed after the cluster is created. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

Step 1

Configure cluster

Step 2

Specify networking

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Configure observability

Step 4

Select add-ons

Step 5

Configure selected add-ons settings

Step 6

Review and create

Specify networking

Networking Info

IP address family and service IP address range cannot be changed after cluster creation.

VPC Info
Select a VPC to use for your EKS cluster resources.

Subnets Info
Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the [VPC console](#).

☒ subnet-06edb215982ca0285
 us-east-1a 172.31.30.0/20

☒ subnet-0d8dcda42ec3ddb7
 us-east-1b 172.31.16.0/20

Additional security groups - optional Info
EKS automatically creates a cluster security group on cluster creation to facilitate communication between worker nodes and control plane. Optionally, choose additional security groups to apply to the EKS-managed Elastic Network Interfaces that are created in your control plane subnets. To create a new security group, go to the corresponding page in the [VPC console](#).

Choose cluster IP address family Info
Specify the IP address type for pods and services in your cluster.

☒ IPv4
 ☐ IPv6

Configure Kubernetes service IP address block Info
☐ Specify the range from which cluster services will receive IP addresses.

Configure remote networks to enable hybrid nodes - new Info
EKS Hybrid Nodes enables you to use on-premises and edge infrastructure as nodes in EKS clusters.

☐ Specify the CIDR blocks for your on-premises environments that you will use for hybrid nodes.

Cluster endpoint access Info

Configure access to the Kubernetes API server endpoint.

☐ Public
 The cluster endpoint is accessible from outside of your VPC. Worker node traffic will leave your VPC to connect to the endpoint.

☒ Public and private
 The cluster endpoint is accessible from outside of your VPC. Worker node traffic to the endpoint will stay within your VPC.

☐ Private
 The cluster endpoint is only accessible through your VPC. Worker node traffic to the endpoint will stay within your VPC.

► Advanced settings

Once it is done, add nodes to cluster. In the cluster options add nodes to Node group.

[EKS](#) > [Clusters](#) > suu-cluster

suu-cluster [Refresh](#) [Delete cluster](#)

Cluster info [Info](#)

Status Active	Kubernetes version Info 1.29	Support type Standard support until March 23, 2025	Provider EKS
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Overview | Resources | **Compute** | Networking | Add-ons | Access | Observability | Upgrade insights | Update history | Tags

Nodes (0) [Info](#)

Node name	Instance type	Node group	Created	Status
No Nodes This cluster does not have any Nodes, or you don't have permission to view them.				

Node groups (0) [Info](#) [Edit](#) [Delete](#) [Add node group](#)

Group name	Desired size	AMI release version	Launch template	Status
No node groups This cluster does not have any node groups. Nodes that are not part of an Amazon EKS managed node group are not shown in the AWS console.				

[Add node group](#)

- Step 1
- Configure node group**
- Step 2
- Set compute and scaling configuration
- Step 3
- Specify networking
- Step 4
- Review and create

Configure node group [Info](#)

A node group is a group of EC2 instances that supply compute capacity to your Amazon EKS cluster. You can add multiple node groups to your cluster.

Node group configuration

These properties cannot be changed after the node group is created.

Name

Assign a unique name for this node group.

suu-node

The node group name should begin with letter or digit and can have any of the following characters: the set of Unicode letters, digits, hyphens and underscores. Maximum length of 63.

Node IAM role [Info](#)

Select the IAM role that will be used by the nodes. To create a new role, go to the [IAM console](#).

LabRole



[Create recommended role](#)

The selected role must not be used by a self-managed node group as this could lead to a service interruption upon managed node group deletion.

[Learn more](#)

Launch template [Info](#)

These properties cannot be changed after the node group is created.

☒ Use launch template

Configure this node group using an EC2 launch template.

- Step 1
- Configure node group
- Step 2
- Set compute and scaling configuration**
- Step 3
- Specify networking
- Step 4
- Review and create

Set compute and scaling configuration

Node group compute configuration

These properties cannot be changed after the node group is created.

AMI type [Info](#)

Select the EKS-optimized Amazon Machine Image for nodes.

Amazon Linux 2023 (x86_64) Standard (AL2023_x86_64_STANDARD)

Capacity type

Select the capacity purchase option for this node group.

On-Demand

Instance types [Info](#)

Select instance types you prefer for this node group.

t3.large

vCPU: 2 vCPUs Memory: 8 GiB Network: Up to 5 Gigabit Max ENI: 3 Max IPs: 36

Disk size

Select the size of the attached EBS volume for each node.

20

GiB

Next, provision the cluster. First setup AWS account on your local machine, please perform the following steps:

1. You can create an environment by typing *aws configure*
 - a. Add at least one data to config

2. Copy your AWS credentials from Vocareum site to local machine. Store them in `~/.aws/credentials`.
3. Create default values for session:
`aws configure set default.region us-east-1`
`aws configure set default.output table`
4. Check the environment (there should be no ec2 instances and no errors on console):
`aws sts get-caller-identity`

Next configure the kubectl command

```
aws eks describe-cluster --region us-east-1 --name <your cluster> --query cluster.status
--output text
aws eks --region us-east-1 update-kubeconfig --name <your cluster>
```

You should see info that the cluster is ready. Now check the status of nodes:

```
kubectl get nodes
```

Also check that all Kubernetes components are ready (here we are using *kube-system* namespace):

```
kubectl --namespace kube-system get pods
```

TASK -1.1 – add value of spent money (e.g. 2\$)

TASK -1.2 – list all nodes and components in your cluster system-wide (same as previous labs)

Task 0 – Create sample application

Based on the provided sample files, create the following deployment and service in a single file:

- Deployment name set to: *kuard-dep*
- use the one single namespace for all resources: *suu*
- set container name to: *kuard*
- set image to *jmutai/kuard-amd64:blue*
- set replicas to *3*
- create service: *kuard-srv*
- set type of service¹ to: *spec.type: LoadBalancer*
- you can separate definitions of a resource in a single file using mark: *---*

You can apply your definitions using the following command:

```
kubectl apply -f .\k8s-2\<your-file>
```

Similar to the previous lab, you can use the “*--dry-run=client -o yaml*” option for debugging. You should be able to access the service using the AWS LB endpoint.

TASK 0 – copy the definition of service and deployment (single file) and add it to the report

Task 1 – Readiness and Liveness endpoints

¹ <https://kubernetes.io/docs/concepts/services-networking/service/>

With pods there are attached readiness and liveness probes². Liveness probe defines if the application is up and running while readiness probe defines if the application can accept user requests. Your pod may be alive but not ready.

Next modify the parameters and deployments. Attach liveness and readiness probe to your deployment and verify if they are working. *Kuard* container have already defined probes under the following URL's: */healthy* and */ready*

Next we will verify the functionality of the probes.

Kuard -dep-041d3b1c3c-q1pww			
Demo application version v0.8.1-1 Serving on 172.31.88.47			
Request Details	Probe is being served on <i>/healthy</i>		
Server Env	Probe will permanently succeed Succeed Fail Fail for next N calls: 1 2 3 5 10		
Memory			
Liveness Probe	ID	When	Status
	99	Apr 21 06:58:04 1 second ago	200
Readiness Probe	98	Apr 21 06:58:01 4 seconds ago	200
	97	Apr 21 06:57:58 7 seconds ago	200
DNS Query	96	Apr 21 06:57:55 10 seconds ago	200
KeyGen Workload	95	Apr 21 06:57:52 13 seconds ago	200
MemQ Server	94	Apr 21 06:57:49 16 seconds ago	200
File system browser	93	Apr 21 06:57:46 19 seconds ago	200
	92	Apr 21 06:57:43 22 seconds ago	200
	91	Apr 21 06:57:40 25 seconds ago	200
	90	Apr 21 06:57:37 28 seconds ago	200
	89	Apr 21 06:57:34 31 seconds ago	200

Using UI set different values for the probes observe the status of pods and containers.

TASK 1 Add your updated descriptor file with both probes defined to the report

Task 2 – Adding jobs

In Kubernetes you can also create short running jobs³. Jobs are short running tasks executed on pods. In the task create job that will send a requests for servers defined in deployment Verify that requests are send and processed by the pods (using *kubect! describe job* and *kubect! logs job/your-job-name*)

TASK 2 – add output of job describe and pod logs to the report

Task 3 – Horizontal Pod Autoscaler

Now, we will use the Horizontal Pod Autoscaler (HPA) to autoscale our deployment.

Use the autoscale command to deploy an HPA. Based on attached HPA configuration add new HPA resource for the deployment.

² <https://kubernetes.io/docs/tasks/configure-pod-container/configure-liveness-readiness-startup-probes/>

³ <https://kubernetes.io/docs/concepts/workloads/controllers/job/>

- Max replicas should be set to **10**, min set to **5**
- CPU utilization should be set to **5%**
- Important: in order to use HPA you need to add resource requirements for your pods⁴

Use the following command

```
kubectrl autoscale deployment kuard-dep --cpu-percent=5 --min=3 --max=10 -n suu
```

Now we can verify the outputs from HPA using:

```
kubectrl get hpa -n suu
```

```
kubectrl describe hpa <name> -n suu
```

Next, modify the job definition from the previous task – add a while loop for constantly querying the service (use the LB endpoint). What happened to the pods?

Add autoscale declaration to your main application deployment description. You can use either *–dry-run* or the attached sample file. In case of API version error, recheck current API support on cluster (*kubectrl api-versions*)

TASK 3 – add output of created hpa object with all events (the key is to show scaling events). Append your latest YAML definition of the environment.

⁴ <https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/>